



Explainable Deep Learning for Machinery Condition Monitoring: Prototype-Based Models vs. Post-Hoc Explanations

Industry Collaboration with Everllence SE (formerly MAN Energy Solutions)

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Abstract

Deep learning performs strongly in machinery fault diagnosis and remaining-useful-life (RUL) estimation [5], but its black-box character blocks adoption in maintenance, where engineers must justify costly interventions. Post-hoc explanations (SHAP, LIME [4]) can be unfaithful to the underlying model [3]; prototype-based networks are inherently interpretable, justifying predictions by similarity to learned, plottable signal patterns [1, 2]. This thesis performs a controlled head-to-head comparison of the two paradigms on standard public benchmarks — CWRU bearing data [8] for fault diagnosis and NASA C-MAPSS [7] for health-stage estimation — measuring both accuracy and explanation faithfulness, complemented by a structured review with Everllence service engineers. The outcome is an evidence-based recommendation on which explanation approach to build into condition-monitoring services, obtained entirely on public data.

Motivation and Gap

Everllence’s engines are continuously instrumented and serviced through the PrimeServ network [9]; converting sensor data into *trusted* maintenance decisions is the practical bottleneck repeatedly identified in the PHM literature [5]. Existing work on explainable predictive maintenance relies almost exclusively on post-hoc explanation, whose faithfulness is contested [3]. What is missing is a controlled comparison, on equal data and matched backbones, between post-hoc explanations and inherently interpretable prototype models [1]: does the interpretability constraint cost accuracy, and are prototype explanations measurably more faithful and more convincing to engineers?

Feasibility

- The workload is bounded and assembled entirely from validated components: standard datasets, established baselines, published architectures, and defined evaluation metrics.
- CWRU and C-MAPSS are the field’s standard benchmarks with published reference pipelines [8, 7] — zero data-acquisition risk; single-GPU compute suffices.
- Prototype architectures have open reference implementations [1], and the author already builds prototype-based self-explainable networks in his current Master’s project.
- No confidential data required; Everllence involvement (expert review) is additive, not blocking.

Research Questions

1. **RQ1:** How does a prototype-based temporal network compare in accuracy to tuned 1D-CNN/LSTM baselines on CWRU and C-MAPSS — how large is the interpretability cost, if any?
2. **RQ2:** Are prototype explanations more faithful than SHAP/saliency explanations of equally accurate black boxes, under deletion/insertion and perturbation metrics?
3. **RQ3:** Which explanation style do Everllence service engineers rate more plausible and actionable in a structured rubric-based review?

Methodology

- **Data/tasks:** CWRU multi-class fault diagnosis (primary); C-MAPSS FD001/FD003 RUL discretised into ordinal health stages (secondary). Published splits; leakage-safe partitioning by unit.
- **Baselines:** 1D-CNN [6] and LSTM tuned to literature-level scores, with SHAP [4] and gradient saliency.
- **Prototype model:** same CNN backbone + ProtoPNet-style prototype layer [1]; prototypes projected onto real training windows; two design variants only (prototype count, window length).
- **Evaluation:** accuracy/macro-F1 and RUL metrics; faithfulness via deletion/insertion and perturbation tests applied uniformly; expert review with 3–5 engineers on anonymised explanation cards, inter-rater agreement reported.
- **Scope/risk:** no deployment or streaming; if the prototype model underperforms, the measured accuracy–interpretability trade-off is itself the (publishable) result; expert review falls back to supervising engineers if availability fails.



Contribution and Value for Everllence

(1) A controlled accuracy comparison quantifying the interpretability cost; (2) a metric-based faithfulness comparison of inherent vs. post-hoc explanations for condition monitoring; (3) open-source pipeline and prototype libraries. For Everllence: a concrete, evidence-based choice of explanation approach for PrimeServ condition-monitoring services — with zero confidential-data dependency.

Timeline

Month	Work
1	Literature review; CWRU/C-MAPSS pipelines; reproduction of baseline reference scores.
2	CNN/LSTM baselines tuned; SHAP and saliency tooling.
3	Prototype model (adapting author's existing codebase); training on CWRU.
4	C-MAPSS health-stage task; full accuracy benchmark (RQ1); faithfulness suite (RQ2).
5	Expert review (RQ3); ablations; result consolidation.
6	Thesis writing, revision, code release, submission.

References

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